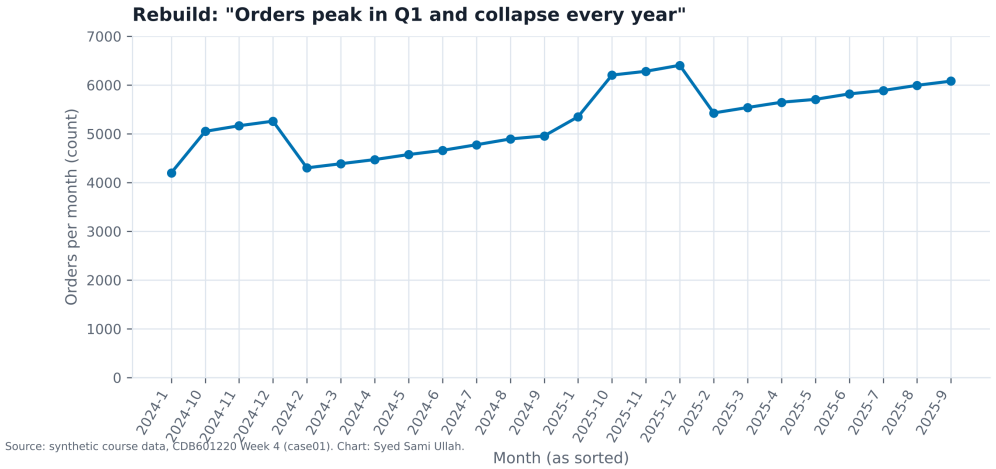
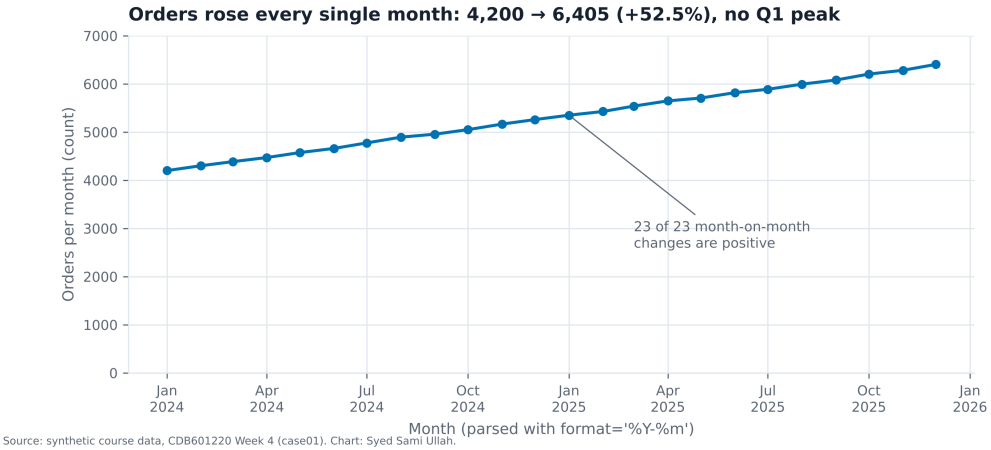
Case 1 · "Orders peak in Q1 and collapse every year - a clear annual cycle"

Recipe: sort the 24 rows by `month`, line with markers, y from zero.





Rebuild (hits the self-check)

Redesign

What this chart shows. Two drops of 958 and 978 orders that look like a yearly crash after Q1. But the x axis reads 2024-1, 2024-10, 2024-11, 2024-12, 2024-2 ... so the months are in the wrong order to begin with.

• **Conclusion it invites:** orders run in a yearly cycle, peaking in Q1 and crashing straight after.

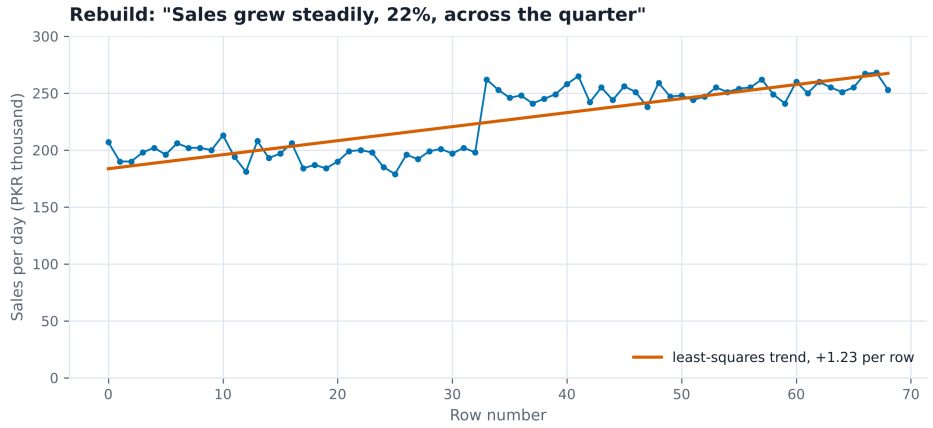
• **Decoy:** my first thought was two branches mixed into one series. Ruled out: `branch` has one value (Gulberg) in all 24 rows.

• **Mechanism:** `month` is text written `YYYY-M` with no zero padding, so the sort is alphabetical. "2024-10", "2024-11", "2024-12" sort before "2024-2", and 22 of the 24 rows end up in the wrong place. Both drops are just December being followed by February: 5,259 → 4,301 (–958) and 6,405 → 5,427 (–978). Parsed as real dates, all 23 monthly changes are positive (smallest +59).

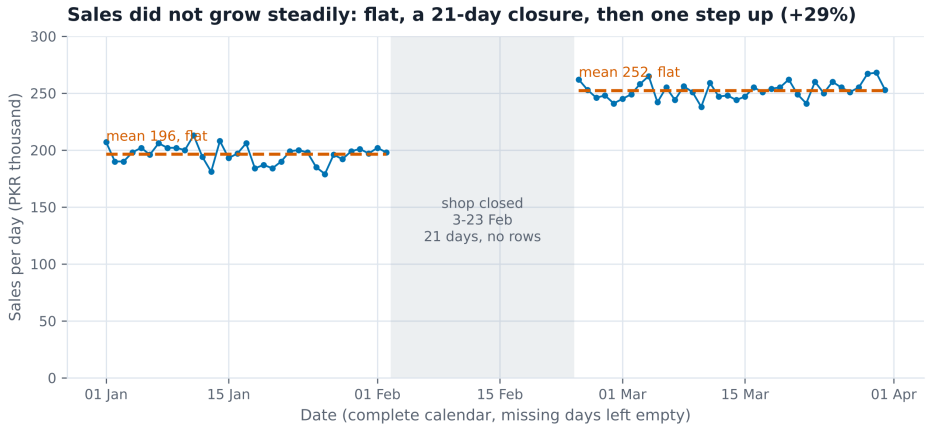
What the redesign shows. With the dates parsed, the "collapse" disappears and it is a plain steady climb. There is no annual cycle in this file at all.

Case 2 · "Sales grew steadily, 22%, across the quarter"

Recipe: plot `sales_pkr_k` against the row number, add a least-squares trend line.



Source: synthetic course data, CDB601220 Week 4 (case02). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case02). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. A smooth upward trend line through 69 points, first 207 and last 253 (+22%). Nothing on it tells you how much time each step on the x axis is.

• **Conclusion it invites:** the shop grew a little every day through the quarter.

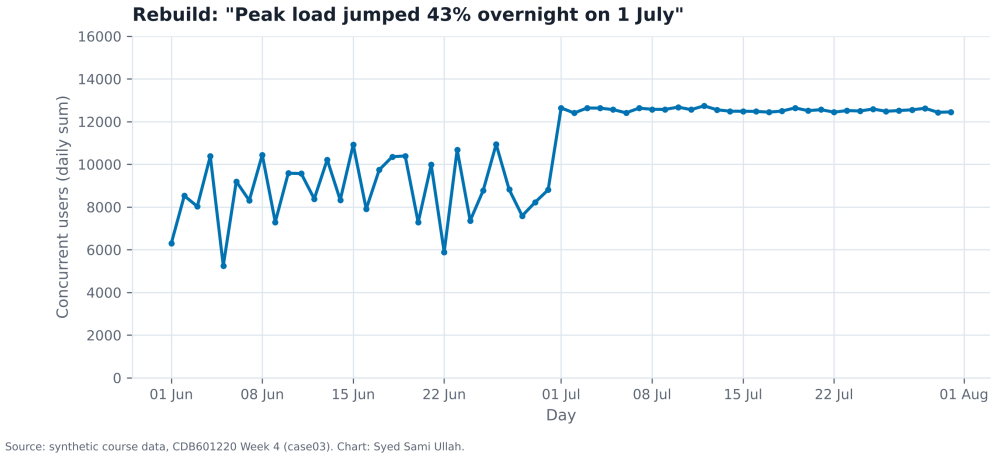
• **Decoy:** "only trading days are recorded" sounds like the weekends are missing, which would be harmless. Ruled out: every weekday appears 9-10 times and the weekday means sit between 222 and 232, so no weekday is closed.

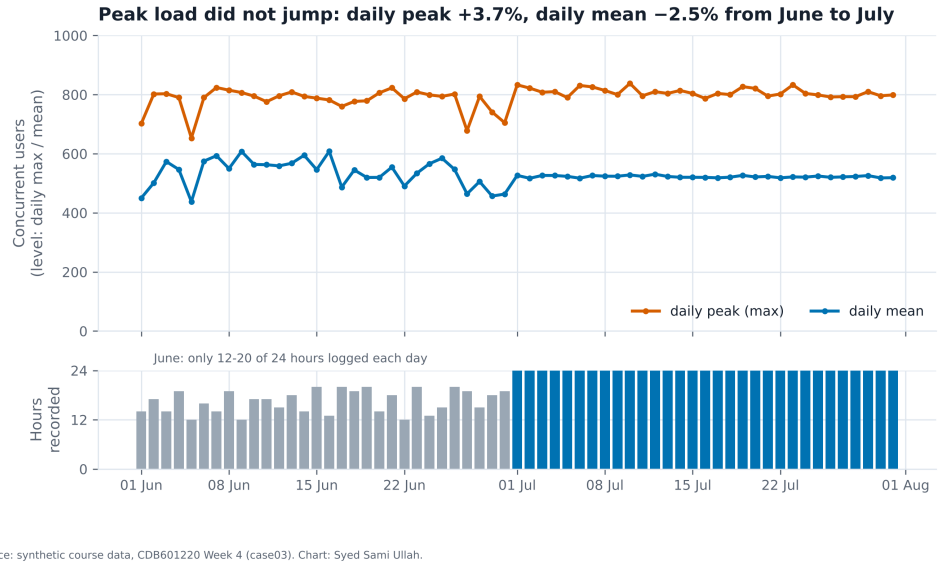
• **Mechanism:** plotting against the row number deletes time. 69 rows cover a 90-day span, and the 21 missing days are one block, 3-23 February, when the shop was shut. Inside each part the trend is flat (–0.16 and +0.23 per day). The whole "+22%" is one step at reopening, 196 → 252 (+28.6%), which the row axis smears into a slope.

What the redesign shows. On a real calendar the closure is visible as a hole, and both sides of it are flat. Whatever caused the jump happened during the closure (a refit, a new menu), not gradually across the quarter.

Case 3 · "Peak load jumped 43% overnight on 1 July"

Recipe: resample `concurrent_users` to daily with `.sum()`, line, y from zero.





Rebuild (hits the self-check)

Redesign

What this chart shows. A clear step from about 8,800 to 12,500 on 1 July. The y label is the giveaway though: a "sum of concurrent users" is not a number of anything.

• **Conclusion it invites:** something on 1 July (a launch, a campaign) pushed peak load up 43% overnight.

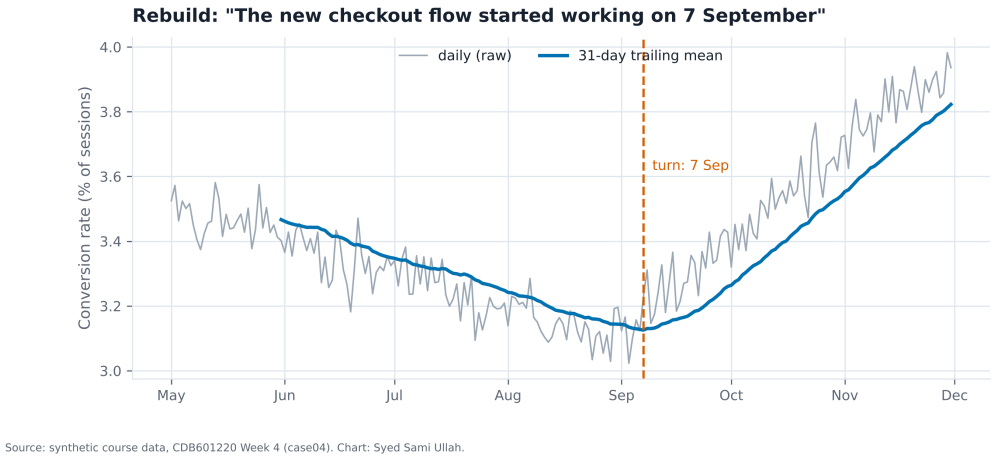
• **Decoy:** `bytes_served` also jumps in July, so the traffic looks real. Ruled out: bytes per *recorded* hour are about the same both months (June 1.06m, July 1.04m, –2.3%), so the flow did not rise either.

• **Mechanism:** `concurrent_users` is a level (users online at an instant) and it was resampled with `sum()`. A sum of a level just counts how many readings there were. June has only 493 of 720 hours (16.4 a day) while July has all 744, and 24 / 16.4 = 1.46, which is the whole "+43%". The proper daily mean actually fell 2.5% and the daily peak rose only 3.7%.

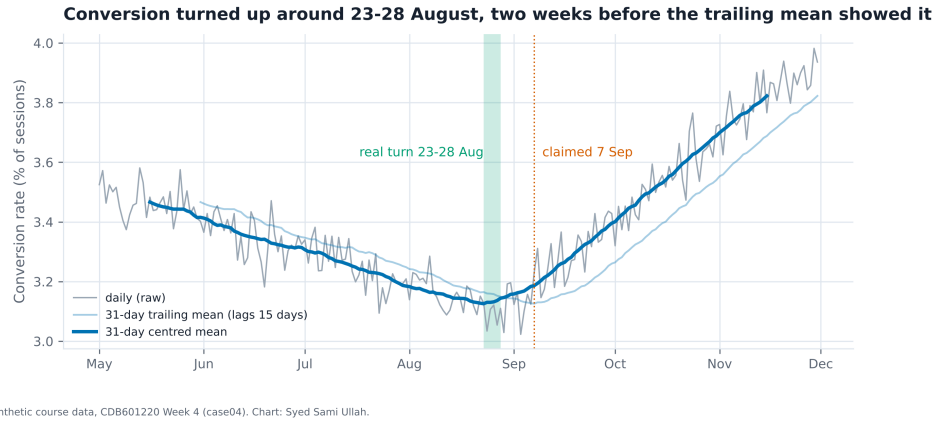
What the redesign shows. Aggregated the way a level should be (mean and max), the load is flat across 1 July. The bottom strip shows the real change on that date: the logger started recording all 24 hours.

Case 4 · "The new checkout flow started working on 7 September"

Recipe: raw `conversion_pct` in grey, 31-day **trailing** mean on top, read the turning point off it.



Source: synthetic course data, CDB601220 Week 4 (case04). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case04). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. The smoothed line bottoms out on 7 September, and the analyst reads that as the day the new flow started helping. The line is drawn at the *end* of each 31-day window, which the chart doesn't say.

• **Conclusion it invites:** the new checkout began lifting conversion on 7 September.

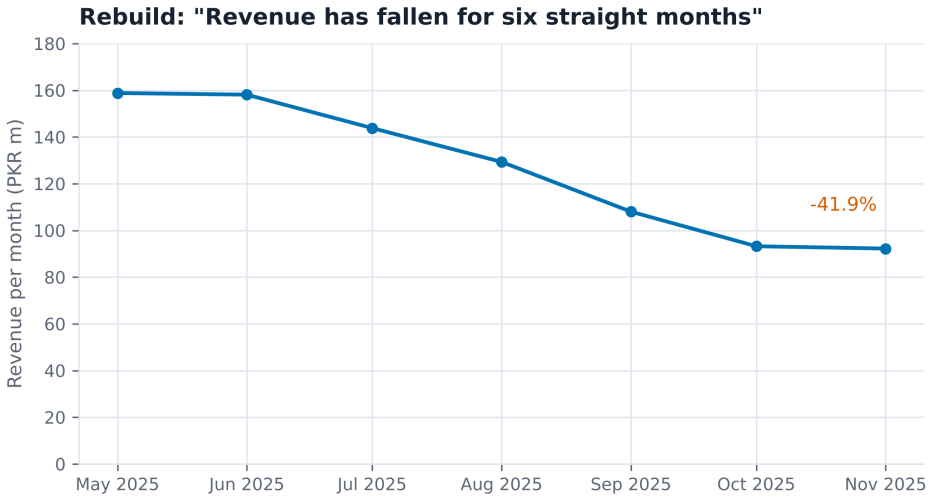
• **Decoy:** "a 31-day mean is so smooth the turn could just be noise." Ruled out: a breakpoint regression on the *raw* data finds a real turn (slope –0.0035 → +0.0092 pp/day). The turn is real, but the date is wrong.

• **Mechanism:** a trailing mean puts each average on the last day of its window, so it cannot see a turn until the window has filled with post-turn days. `lag_at_turn` gives 15 days, exactly half the window: centred minimum 23 August, trailing minimum 7 September. The raw hinge fit puts the break on 28 August. Either way, the flow started working in the last week of August, about two weeks before the date on the chart.

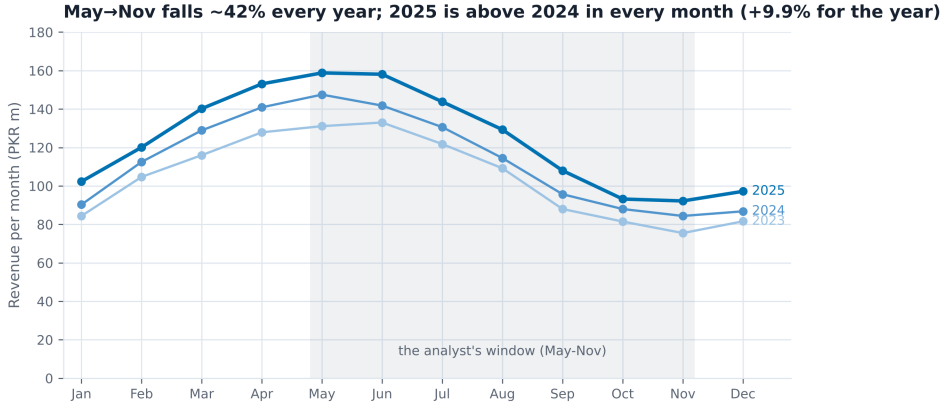
What the redesign shows. The centred mean and the raw breakpoint agree on late August. If the new checkout went live on 7 September it cannot be the cause, so someone needs to check what changed in late August.

Case 5 · "Revenue has fallen for six straight months"

Recipe: May to November 2025 only, line with markers, y from zero.



Source: synthetic course data, CDB601220 Week 4 (case05). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case05). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. Six falls in a row, 158.8 down to 92.2, on an honest zero-based axis. Every point is correct; what's wrong is where the window starts and stops.

• **Conclusion it invites:** the business is in decline.

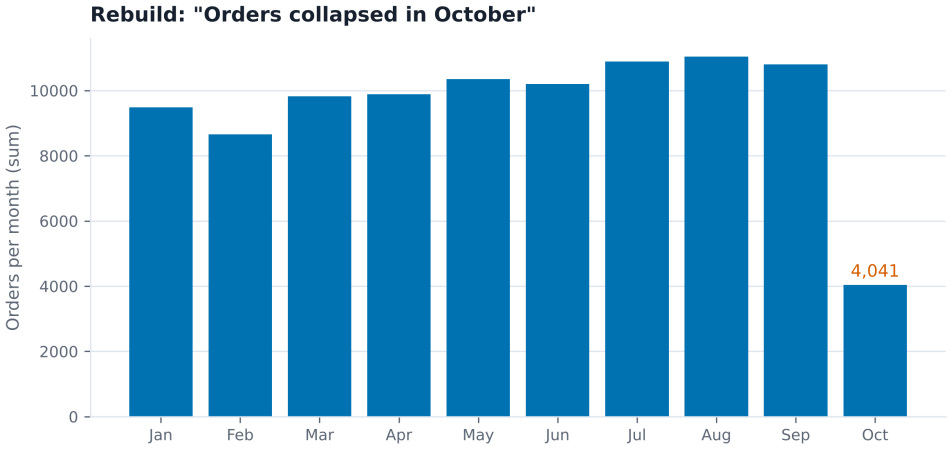
• **Decoy:** the brief warns against two. The axis is not truncated (it starts at 0), and the sample is not small: 36 complete months exist, the analyst simply used 7 of them.

• **Mechanism:** the window starts at the seasonal peak and ends at the trough. November is the trough in all three years and May is the peak in 2024 and 2025 (June in 2023, by 1.9m), and May → Nov fell –42.4% in 2023 and –42.8% in 2024 too. Against the same months of 2024, every month in the window is *up* (+5.9% to +12.9%), and 2025 as a whole is up 9.9%.

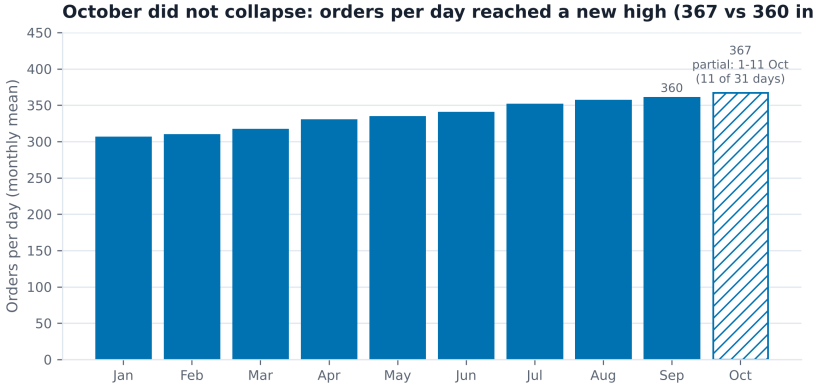
What the redesign shows. Same line, same zero axis, with the earlier years next to it. The "six-month fall" is just the usual seasonal slide, and 2025 is the highest of the three lines the whole way through. This is one of the two cases that was incomplete rather than wrong.

Case 6 · "Orders collapsed in October"

Recipe: monthly totals with `.sum()`, bars from zero.



Source: synthetic course data, CDB601220 Week 4 (case06). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case06). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. Nine bars around 9-11k and then October at 4,041, about 37% of September. A drop like that looks like a crisis.

• **Conclusion it invites:** demand fell off a cliff in October.

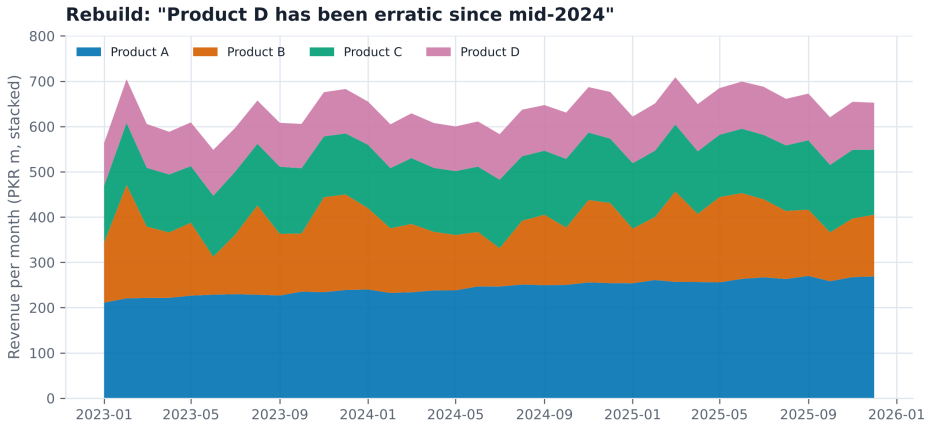
• **Decoy:** maybe a real outage or holiday in October. Ruled out: there are no zero days, the lowest October day is 347, and October's daily average (367.4) is *higher* than September's (360.2).

• **Mechanism:** an incomplete final period. The extract stops on 11 October (`audit_index` flags `partial_tail`, 11 rows), so October holds 11 of its 31 days. 11/31 = 35.5%, and with the slightly higher daily rate that gives the 37%. At its own pace October would be about 11,390 orders, a record.

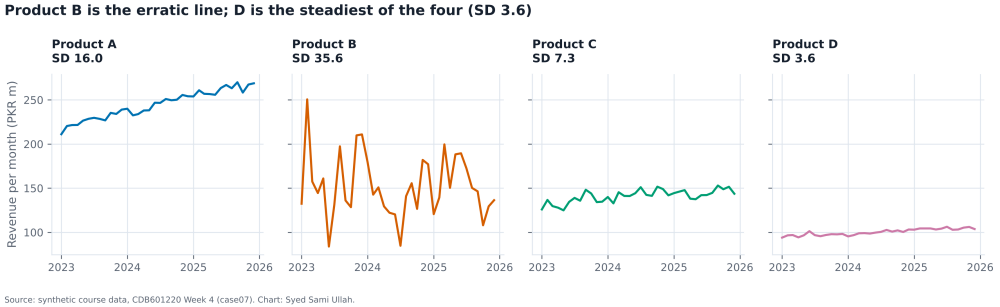
What the redesign shows. Switching to orders per day makes months of different length comparable, and the hatched October bar is marked as partial. The trend is a steady climb all year, and October continues it.

Case 7 · "Product D has been erratic since mid-2024"

Recipe: stacked area of the four products, `product_a` at the bottom, `product_d` on top.



Source: synthetic course data, CDB601220 Week 4 (case07). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case07). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. The top band (D) has the most jagged edge on the chart, so it reads as the unstable product. But the top edge of D is the total of all four, not D.

• **Conclusion it invites:** product D became unstable from mid-2024 and needs looking into.

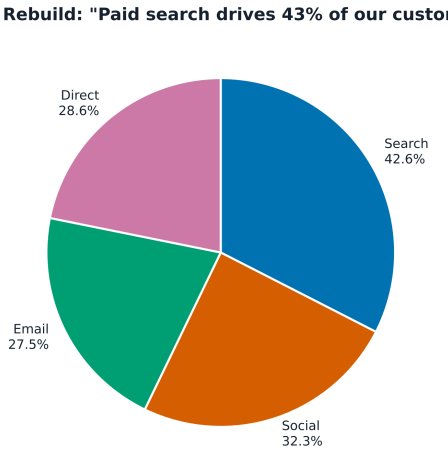
• **Decoy:** something did change for D in mid-2024. Ruled out: D's month-to-month change SD is 2.2 before July 2024 and 1.8 after, and it never leaves 94-106. It is the steadiest of the four products.

• **Mechanism:** in a stack each band sits on a moving baseline (A+B+C), so D's band inherits everyone else's wobble. The column the recipe never isolates is `product_b`: SD 35.6, monthly change SD 45.8, swinging between 83.9 and 250.6. The top edge correlates 0.79 with B and only 0.43 with D itself.

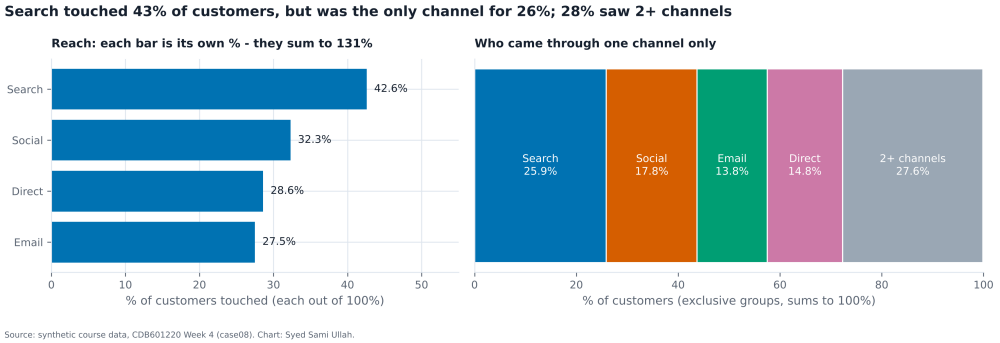
What the redesign shows. Unstacked into small multiples, each product sits on its own flat baseline with a shared y scale. D is an almost straight line, and all the noise belongs to B.

Case 8 · "Paid search drives 43% of our customers"

Recipe: count `True` in each `touch_*` column, divide by customers, four-slice pie.



Source: synthetic course data, CDB601220 Week 4 (case08). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case08). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. Search looks like the biggest slice and its label says 42.6%. Add the four labels and you get 131%, so this can't be a whole.

• **Conclusion it invites:** search is the channel behind 43% of customers, so it deserves the biggest budget.

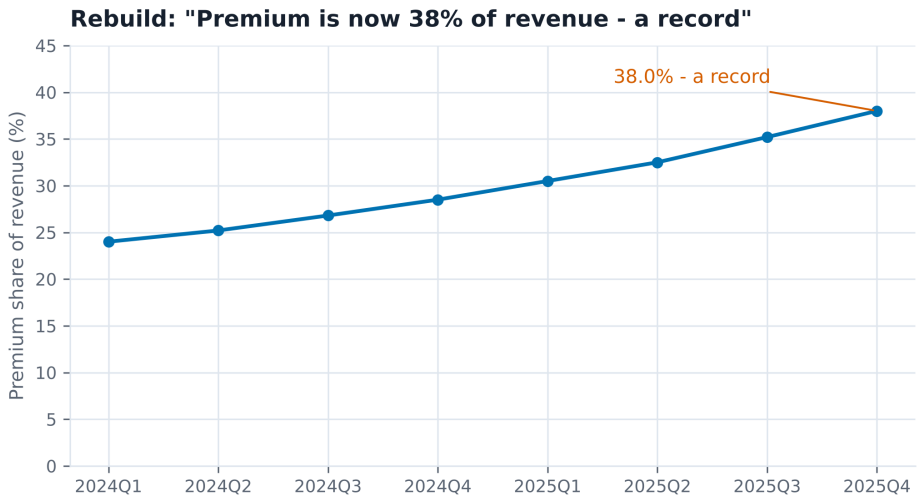
• **Decoy:** maybe customers are double-counted as duplicate rows. Ruled out: 4,000 rows and 4,000 unique `customer_ids`.

• **Mechanism:** the parts are not exclusive. 1,105 customers (27.6%) were touched by two or more channels, so the four shares sum to 131.0%. A pie silently re-normalises, so search's wedge is really 32.5% of the circle while its label says 42.6%. "Touched" is also not "drove": only 25.9% of customers came through search alone.

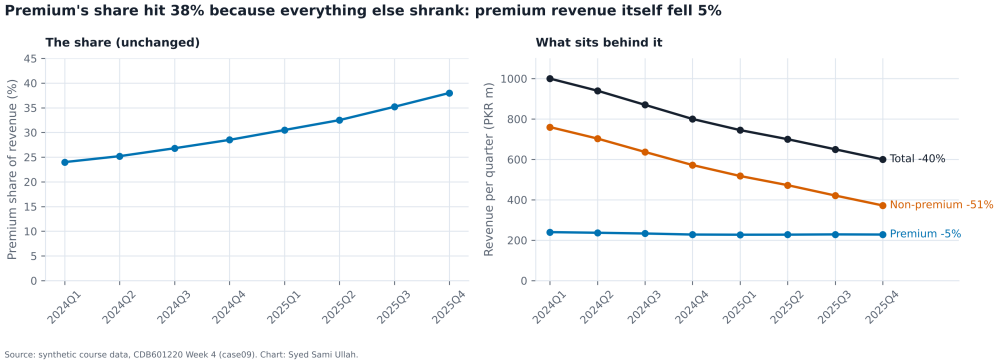
What the redesign shows. Reach is shown as separate bars (not a whole), and the real part-to-whole is the exclusive split on the right, which adds to 100%. Search is still the largest single channel, but at 26%, not 43%.

Case 9 · "Premium is now 38% of revenue - a record"

Recipe: premium / total per quarter, line, y from zero, annotate the last point.



Source: synthetic course data, CDB601220 Week 4 (case09). Chart: Syed Sami Ullah.



Source: synthetic course data, CDB601220 Week 4 (case09). Chart: Syed Sami Ullah.

Rebuild (hits the self-check)

Redesign

What this chart shows. A clean rise from 24% to 38% over eight quarters. A share is a ratio, and the chart gives no hint whether the top or the bottom of the ratio moved.

• **Conclusion it invites:** premium is booming and should get more investment.

• **Decoy:** maybe the share is computed wrong or the parts don't fit the whole. Ruled out: premium is below total in every quarter and "other" is positive, so the share is arithmetically correct.

• **Mechanism:** a rising share of a falling total. Total revenue fell 40% (1,000 → 600) and the non-premium part fell 51% (760 → 372). Premium revenue itself went *down* 5% (240 → 228). The record share is entirely the denominator shrinking.

What the redesign shows. The same share line, with the absolute revenue next to it. Premium is flat-to-falling while total revenue lost 40%, which is a much worse story than "a record". This is the second incomplete-not-wrong case.